1

Lect No 1

Applied physics

What is Physics?

- Natural philosphy
  - → Aristotle
  - → Galio-Galillie
  - → Biological Science
    - ↳ study of life
  - → Physical science
    - ↳ Non-living things

Science

Knowledge gained through observation and experiment.

Physical Science

- → Geology
  - ↳ study of Earth
- → Astronomy
  - ↳ study of stars
- → Chemistry
  - ↳ chemicals
- → Physics

Physics:

Study of matter and energy and their relationship. / Physics is study of universe.

Matter:

- ↳ have mass
- ↳ have volume

Types

- Solid
- Liquid
- Sound
- ~~Heat~~ Plasma

Energy:

- Light
- Sound
- Heat.

Relationship:- $E = mc^2$

Atom Bomb:

Uranium ⟶ Energy

Pair production:

Energy → Matter

Physics

- Classical Physics (17th century before)
- Modern Physics (19th/20th)

cant explain by classical physics

- → Newton
  - Mechanics
  - Thermodynamics
  - Electromegnatism
- → Black body Radition
  - Einstein
  - Max planx
- → Quantum Physics
- → Nuclear Physics
- → Particle Physics.

Electromegnatism

$$\text{Electromegnatism} \rightarrow \text{Electricity},\quad \text{Magnetism}$$

Electricity
↓
Phenomenon arises due to motion of electrical charge
$I = Q/t$

Magnetism
↓
Phenomenon by which material attract or repell metallic substance.

Maxwell:-
→ combine E·f and M·f
Maxwell equation:
Relationship b/w E·f and M·f.

- **Electrical charge:**
  Electrical charge is intrinsic property of property of matter, associated with its elementry particles due to which it produce and experience electric and Magnetic effects.

~~cant explain by~~
classical physics.

→ Newtom
- Mechanics
- Thermodynamics
- Electromegnatism

→ Black body Radition
- Einstein
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→ Quantum Physics
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Electromegnatism

Electricity
Phenomenon aries due to motion of electrical charge
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Phenomenon by which material attract or repell metallic substance.

Maxwell:-
→ combine E·F and M·f
Maxwell equation:
Relationship b/w E·f and M·f.

- Electrical charge:
Electrical charge is intrinsic property of property of matter, associated with its elementry particles due to which it produce and experience electric and Magnetic effects.

• Mass is also intrinsic property

→ Mass can exist without charge
Neutron = 0 charge
$= 1.67 \times 10^{-27}$ kg

→ charge cannot exist without mass.
electron = -1.6×10^{-19} C
$= 9.1 \times 10^{-31}$ kg
Proton = $+1.6 \times 10^{-19}$ C
$= 1.6 \times 10^{-27}$ kg

• Types of charge:
Beajimne franklin Divided charges into two types.
→ Positive charge (Proton)
→ Negative charge (electron)

It is produced by gain of electron

It is produced by removing electron.

→ Charge is Scalar quantity (Columbs)
SI unit C = As

esu emu	static columb

• Point charge
→ Charge on a point.
→ Dimension less charge.
→ Having a large distance between them as compared to their size.

• Basic laws of Electrostatic:
→ Like charges repell each other.
→ Unlike charges attract each oher.
→ A charge body always attract a neutral body. (Because of creation of dipole)

Lec No 2

→ **Smallest charge:**

Smallest independent charge that can exist is charge or electron or charge on proton.

$1e \rightarrow -1.6 \times 10^{-19} C \rightarrow -e$

$1p \rightarrow +1.6 \times 10^{-19} C \rightarrow +e$

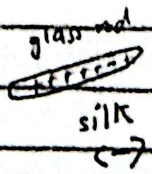
• **Quarks**

→ Up $+2/3\,e$
→ Down $-1/3\,e$
→ Strange $-1/3\,e$
→ charm $+2/3\,e$
→ Top $+2/3\,e$
→ Bottom $-1/3\,e$

→ Mass only attract.
→ Charge can attract or repell

A Body can by charged by

* By friction:



Plastic (-ve)
Fur (+ve)

Two natural bodies are rubed together.

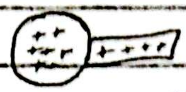
Dont know the reason of -ve and +ve charge on body. It vary from body to body.

2 - 5
- 3

* By conduction:

A charge body is brought in contact with neutral body

(N)



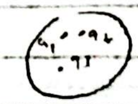
after contact charge flow from +ve to neutral body until it become balance.

* By induction:

A charge body is brought near to neutral body.

Properties of electric charge

(i) Additive property.



Consider a system of charges having three points charges with magnitude q_1, q_2, q_3, then total charge is

$$Q_t = q_1 + q_2 + q_3$$

In case of n

$$Q_t = q_1 + q_2 + q_3 + \cdots + q_n$$

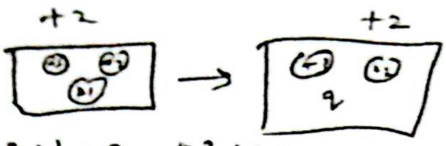
ii) Conservation of charge:

Total charge of an isolated system remains constant over time.

Algebric sum of charge in a close system always remain same.

$$\Sigma Q = \text{constant}$$

net initial charge = net ~~final charge~~



same net charge.

$+3+1-2 = +3+2+q$

$q = -3$

* $n^{\circ} \longrightarrow p^{+1} + e^{-e}$

* $e^{-1} + e^{+1} \longrightarrow \gamma + \gamma$ (gema-rays)

(fix value)

iii) Quantization of charge:

* A body can only have charge multiple of e. (Integral multiple of e)
* When we transfer electric charge from one object to another in form of electric current, then this flow of electric charge is not continuous, It will only have discrete values.
* Electric charge cannot be divided into small parts.
* $q = ne \quad \because n = \pm 1, \pm 2, \pm 3, \ldots$

  It means that charge only exist in quantities that are integral multiple of certain elementry Quantity. of e

Quantitative analysis of charge

* Coulomb's Law (1784, French militry engineer)
  - → Emperical's Law (First experiment then theory)
  - → Inverse square law
  - → Only applicable on point charge. and static charges.
  - → Charges must be far enough so that strong nuclear force does not dominate coulomb's force.

CamScanner

Coulomb's torisons Balance

(2)

statement:

Electrostatic force of attraction or repulsion is directly proportional to the product of magnitude of charge and inversly proportional to square of distance b/w them.

$$F \propto \frac{|q_1||q_2|}{r^2}$$

$$F = \frac{K |q_1||q_2|}{r^2}$$

K is coulomb's constant

$$K = \frac{Fr^2}{|q_1||q_2|}$$

K units - Nm^2C^{-2}

K value depend on

* Nature of medium b/w charges
* System of units.

In vaccum

$$F_{vac} = K_{vac} \frac{|q_1||q_2|}{r^2}$$

$$K = \frac{1}{4\pi \varepsilon_0} = 9 \times 10^9 \, Nm^2C^{-2}$$

$\varepsilon_0 = 8.85 \times 10^{-12} N^{-1} m^{-2} C^2$ Permitivity of free space

$\varepsilon_0 \downarrow \quad F \uparrow$

$$F_{med} = K_{med} \frac{|q_1||q_2|}{r^2}$$

k , = 1 ε , , ε ε

$$F_{med} = \frac{1}{4\pi\varepsilon_0\varepsilon_r} \frac{|q_1||q_2|}{r^2}$$

$$F_{med} = \frac{F_{vac}}{\varepsilon_r} \qquad \downarrow F_{med} \propto \frac{1}{\varepsilon_r \uparrow}$$

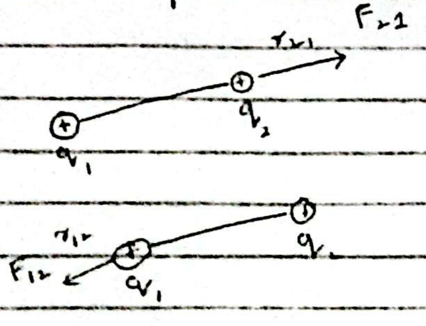
$$F_{med} < F_{vacuum}$$

* Presence of dielectric reduces the coulomb force by a factor known as ε_r.

* Vector form:

$$F = \frac{1}{4\pi\varepsilon_0} \frac{|q_1||q_2|}{r^2} \hat{r}$$

* Mutual force:



$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0} \frac{|q_1||q_2|}{r^2} \hat{r}_{21}$$

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{|q_1||q_2|}{r^2} \hat{r}_{12}$$

By newton 3rd Law:- $\hat{r}_{21} = -\hat{r}_{12}$

$$\vec{F}_{21} = -\vec{F}_{12}$$

3

Lecture # 3 :

Distribution of Charge { Uniform distribution/continuous (U) ; non-uniform distribution/discrete (D)

→ • Uniform distribution
  * Linear distribution of charge
  * Surface distribution of charge
  * Volume distribution of charge → hollow

* Linear :
  • Linear Charge density = Charge per unit length

$$\lambda = \frac{q}{l} = \frac{C}{m}$$

Unit = Cm^{-1}

Conducting sphere

* Surface :
  • Surface Charge density : Charge per unit Area

$$\sigma = \frac{q}{A} = Cm^{-2}$$

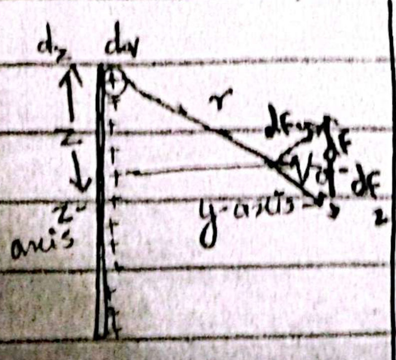
→ non-conducting → charge will remain inside

* Volume :
  • Volume Charge density : Charge per unit volume.

$$\delta = \frac{q}{V} = Cm^{-3}$$

A. Example (Application) of linear.

. A uniform line of Charge:



$dyz = 0$

Consider a thin Rod of length $-L/2$ to $+L/2$ that lies along z-axis & carries a uniformly distributed +ve charge (q). So, linear charge density :

$$\lambda = \frac{q}{L}$$

We want to calculate force exerted by rod on a point charge (+ve) q_0, placed at perpendicular bisector of rod at distance "y".

We divided rod into small elements of length d_z

An arbitrary charge "d_q" is considered such as it is at distance "z" from centre of rod

$$\lambda = \frac{dq}{dz}$$

$$dq = \lambda dz$$

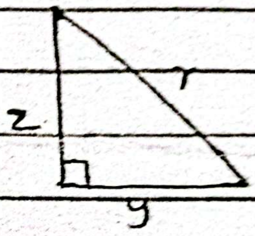
$$dF = \frac{kq_0\,dq}{r^2} \longrightarrow (1)$$

As z-axis forces are antiparallel so net is zero

Along y-axis,

$$dF_y = dF\cos\theta \longrightarrow (2)$$

$$dF_y = \frac{kq_0 dq}{r^2}(\cos\theta) \longrightarrow (3)$$



$$dF_y = \frac{kq_0\,dq}{r^2}\left(\frac{y}{r}\right) \qquad \because \cos\theta = \frac{base}{hyp}$$

$$dF_y = \frac{kq_0\,dq\,y}{r^3}$$

$$dF_y = \frac{kq_0\lambda\,dz\,y}{r^3}$$

$$\int dF_y = kq_0\lambda y \int_{-L/2}^{+L/2} \frac{dz}{r^3} \quad \text{length}$$

$$F_y = kq_0\lambda y \int_{-L/2}^{+L/2} \frac{dz}{(y^2+z^2)^{3/2}}$$

$$\int_{-L/2}^{+L/2} \frac{d_z}{(y^2+z^2)^{3/2}}$$

integration by substitution

$$\boxed{\begin{aligned} z &= y\tan\theta \\ dz &= y\sec^2\theta \cdot d\theta \end{aligned}}$$

$z = y\tan\theta$

$\frac{-L}{2} = y\tan\theta$

$\theta = -\tan^{-1}\left(\frac{+L}{2y}\right)$

$z = y\tan\theta$

$\frac{L}{2} = y\tan\theta$

$\theta = \tan^{-1}\left(\frac{L}{2y}\right)$

$\frac{1}{\sec\theta} = \cos\theta$

$$\int_{-\tan^{-1}\left(\frac{L}{2y}\right)}^{\tan^{-1}\left(\frac{L}{2y}\right)} \frac{y\sec^2\theta \cdot d\theta}{(y^2+y^2\tan^2\theta)^{3/2}}$$

$-x$ $+x$

$$\int_{-x}^{x} \frac{y\sec^2\theta \cdot d\theta}{y^{\cancel{2}\times 3/\cancel{2}}(1+\tan^2\theta)^{3/2}}$$

$$\int_{-x}^{x} \frac{y\sec^2\theta \cdot d\theta}{y^{3}(\sec^2\theta)^{3/2}}$$

$$\int_{-x}^{x} \frac{1 \cdot d\theta}{y^2\sec\theta}$$

$$\frac{1}{y^2}\int_{-x}^{x} \cos\theta \qquad \therefore \frac{1}{\sec\theta} = \cos\theta$$

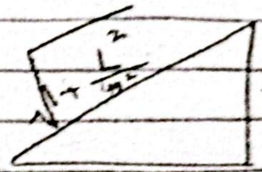
$$\frac{1}{y^2}\sin\theta \Big|_{-x}^{+x} \qquad \therefore \int\cos\theta = \sin\theta$$

$$\frac{1}{y^2}\sin(x) - \sin(-x)$$

$$\boxed{\frac{2}{y^2}\sin x}$$

$$\frac{2}{y^2}\sin\left(\tan^{-1}\frac{L}{2y}\right)$$

$$\tan\theta = \frac{L}{2y}$$



$\sqrt{1+\frac{L^2}{4y^2}}$

$\frac{L}{2y} = P$

$1 = B$

~~hyp~~ =

$\therefore \sin = \frac{P}{h}$

$$\frac{2}{y^2}\times\frac{L/2y}{\sqrt{1+L^2/4y^2}}$$

$$\frac{\cancel{2}}{y^2}\times\frac{L}{\cancel{2}y}\times\frac{1}{\sqrt{1+\frac{L^2}{4y^2}}}$$

$$\frac{L}{y^3}\times\frac{1}{\sqrt{1+\frac{L^2}{4y^2}}}$$

$$\frac{L}{y^3}\times\frac{1}{\sqrt{\frac{1}{y^2}\left(y^2+\frac{L^2y^2}{4y^2}\right)}}$$

$$\frac{L}{y^{\cancel{3}2}}\times\frac{1}{\frac{1}{\cancel{y}}\sqrt{y^2+\frac{L^2}{4}}}$$

$$\frac{L}{y^2}\times\frac{1}{\sqrt{y^2+\frac{L^2}{4}}}$$

$$F_y = kq_0\lambda y\int_{-L/2}^{+L/2}\frac{dz}{(y^2+z^2)^{3/2}}$$

$$F_y = kq_0\left(\frac{q}{\cancel{L}}\right)\cancel{y}\times\frac{\cancel{L}}{y^2}\times\frac{1}{\sqrt{y^2+\frac{L^2}{4}}}$$

$$F_y = kq_0q_0\times\frac{1}{y\times\sqrt{y^2+\frac{2^2}{4}}}$$

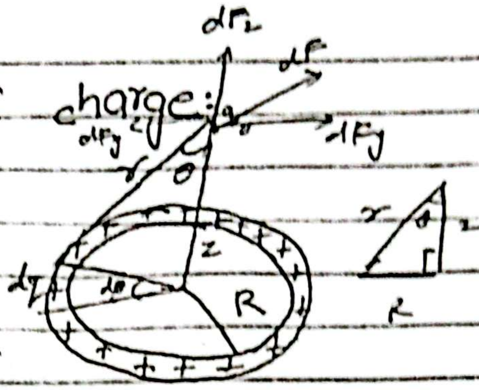
$$\boxed{F_y = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_0q}{y\sqrt{y^2+\frac{L^2}{4}}}}$$

$$E = \frac{F}{q_0}$$

$$\boxed{E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{y\sqrt{y^2+\frac{L^2}{4}}}}$$

④ Lecture No 4

★ Force due to ring of charge:



Consider a ring of radius 'R' carrying a uniformly distributed (+ve) charge 'q' such that Linear charge density will be

$$\lambda = \frac{q}{2\pi r}$$

$s = r\theta$

$l = R\,d\varphi$

→ We want to calculate the force exerted by ring on a charge q_0, lies along z-axis at a distance z from the centre of ring.

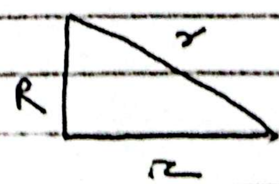
Consider a small Lenght $Rd\varphi$, having charge dq

$$\lambda = \frac{dq}{Rd\varphi}$$

$$dq = \lambda R d\varphi$$

$$dF = \frac{Kq_0\,dq}{r^2} \quad (A)$$

$$dF_z = dF.\cos\theta \quad (B)$$



$$dF_z = \frac{Kq_0\,dq}{r^2}\left(\frac{z}{r}\right)$$

$$r = \sqrt{z^2 + R^2}$$

$$dF_z = \frac{Kq_0\,\lambda R d\varphi \,.\, z}{r^3}$$

*

$$dE_z = \frac{Kq_0 (\lambda R d\varphi) \cdot z}{(z^2+R^2)^{3/2}}$$

$$dE_z = \frac{Kq_0 \lambda R z}{(z^2+R^2)^{3/2}} d\varphi$$

$$\int \cancel{d}E_z = \frac{Kq_0 (R\lambda z)}{(z^2+R^2)^{3/2}} \int_0^{2\pi} d\varphi$$

$$E_z = \frac{Kq_0 (R\lambda z)}{(z^2+R^2)^{3/2}} \cdot \varphi \Big|_0^{2\pi}$$

$$= \frac{Kq_0 (R\lambda z)}{(z^2+R^2)^{3/2}} (2\pi)$$

$$= \frac{Kq_0 \left(q/\cancel{2\pi R} \right) z (\cancel{2\pi R})}{(z^2+R^2)^{3/2}}$$

$$E_z = \frac{Kq_0 q z}{(z^2+R^2)^{3/2}}$$

$$E = F/q_0$$

$$E = \frac{Kqz}{(z^2+R^2)^{3/2}}$$

id free region or neutral zone

* $z >>> R$

$R = 0$

$$F_z = \frac{kq_0 q z}{(z^2)^{3/2}}$$

$$= \frac{kq_0 q z}{z^3}$$

$$F_z = \frac{kq_0 q}{z^2}$$

Them force will become equal to coulombs force.

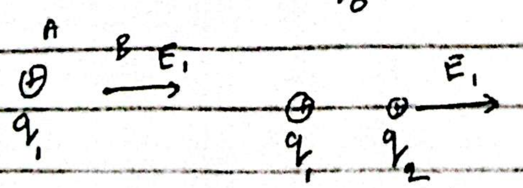
case 2:

* $z = 0$

So $F_z = 0$

Because at the centre of ring charge (q_0) will be pushed by all the charge element present on the Ring.

$\rightarrow E = \lim_{q_0 \to 0} \frac{F}{q_0}$



$F_{21} = q_2 E_1$

$$aF_2 = \frac{\Lambda r_0 \ldots}{r^3}$$

* Electric field:

Region or space around a charge in which it can exert electrostatic force of attraction or repulsion on another charge.

* Electric field intensity:

If a small charge (q_0) will experience a force 'F' in electric field E then electric field intensity is

$$\vec{E} = \frac{\vec{F}}{q_0}$$

→ It is a vector quantity

→ It's ~~is~~ unit is N/C

→ It is always in direction F.

→ q_0 must be small charge so that it will not disturb the field Produced by source charges.

5

Lect No 5

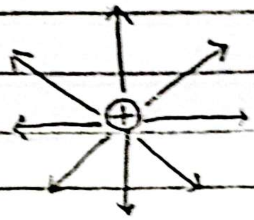
$$F_{12} = q_1 E_2$$

$$F_{21} = -F_{12}$$

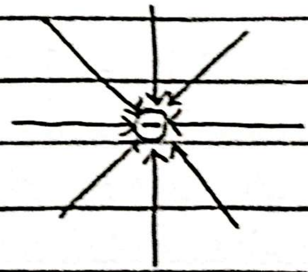
* Electric field lines:

Electric field lines are drawn to describe the direction of E.F around a charge.

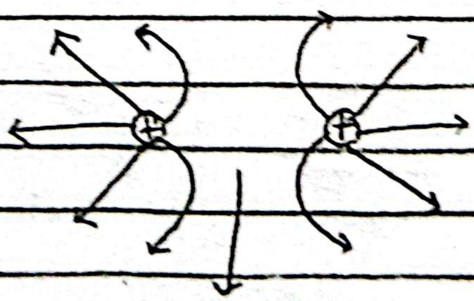
(i) E.F lines due to +ve charge:



(ii) E.F lines due to -ve charge:

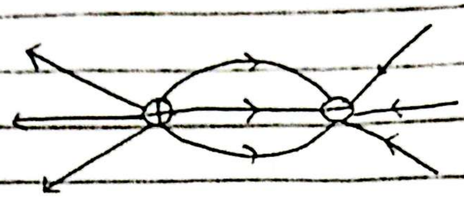


(iii) E.F lines due to same charge:

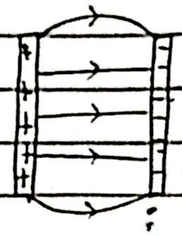


field free region or neutral zone

(iv) E.F lines due to opposite charge:



(v) E.F lines b/w opposite charge plates:



* Properties:

→ Electric field lines never cross each other. Because E.F lines have only single direction at given point.

→ Electric field lines orignate from +ve charge, and terminate at negative charge.

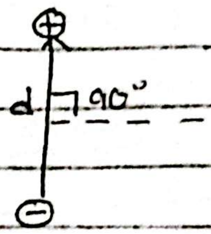
→ Tagent to field line at a point given the direction E.F at that point.

→ Electric field is strong where field lines are ~~strong~~ close and where weak field lines are further apart from each other.

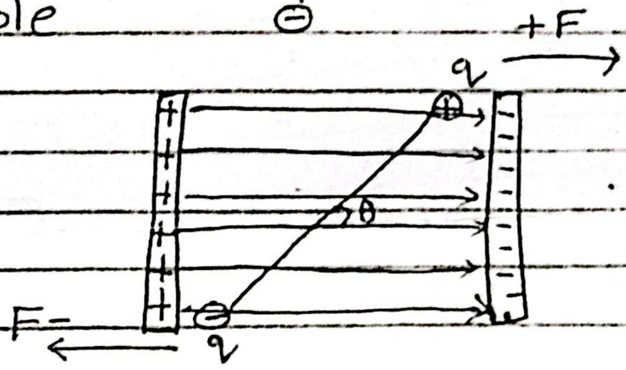
→ Electric field lines are imaginary But electric field is real.

* Dipole In Electric field:

→ A pair of +ve and -ve charge separated by a constant distance 'd' form a dipole



$P = q\,d$
↙ Dipole moment

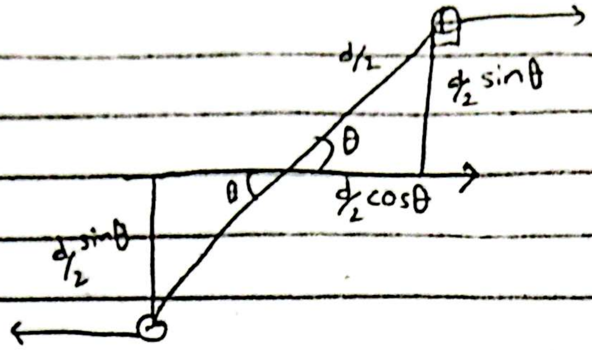


→ Let this dipole is placed in an external E·F, such that its electric dipole moment vector $\vec{P}$ maked angle 'θ' with '$\vec{E}$'.

→ Let Force (+F) will be exerted on (+q) and (-F) will be exerted on (-q).

→ As the forces are equal in magnitude but opposite in direction, so, they will produce couple and this couple will produce a torque, which will tend to rotate dipole around it centre of mass.

$|+F| = |-F| = F$



$$\tau = \tau_1 + \tau_2$$
$$= \left(\frac{d}{2}\sin\theta\right)F + \left(\frac{d}{2}\sin\theta\right)F$$
$$= \frac{d}{2}F\sin\theta + \frac{d}{2}F\sin\theta$$
$$= F\sin\theta\left(\frac{d}{2} + \frac{d}{2}\right)$$
$$= F\sin\theta\left(\frac{\cancel{2}d}{\cancel{2}}\right)$$
$$\tau = Fd\sin\theta \qquad \because F = qE$$
$$= (qE)d\sin\theta$$
$$= (qd)E\sin\theta$$
$$\tau = PE\sin\theta \qquad \because P = qd$$
$$= \vec{P} \times \vec{E}$$

By right hand rule its direction will be into the plane of the board.

*** Work done:**

$$dw = F \cdot ds \qquad \because s = r\theta$$
$$dw = F \cdot d(r\theta)$$
$$= rF \cdot d\theta$$
$$dw = \tau \cdot d\theta$$

$\rightarrow d\theta$ is angular displacement.

$$\int dW = -\int_{\theta_0}^{\theta} \tau \cdot d\theta$$

negative sign is showing that angle decrease.

$$W = -\int_{\theta_0}^{\theta} \tau \cdot d\theta$$

$$W = -\int_{\theta_0}^{\theta} PE\sin\theta \cdot d\theta$$

$$W = -PE(-\cos\theta)\Big|_{\theta_0}^{\theta}$$

$$W = +PE(\cos\theta - \cos\theta_0)$$

*** Energy:**

$$\Delta U = -W$$

$$= -PE(\cos\theta - \cos\theta_0)$$

initial angle $90° = \theta_0$

$$U(\theta) - U(\theta_0) = -PE(\cos\theta - 0)$$

$$U(\theta) = -PE\cos\theta$$

*** Max P.E:**

$\vec{P}$ and $\vec{E}$ are antiparallel

$$U(\theta) = -PE\cos(180)$$

$$= -PE\cos(-1)$$

$$U(\theta) = PE$$

*** Min P.E:**

P are E are parallel $U(\theta) = -PE\cos 0°$

$$U(\theta) = -PE(1)$$

$$U\theta = -PE$$

Lect No 5

Lecture 05:

Electric Field:

Region or space around a charge in which it can exert electrostatic force of attraction or repulsion on another charge.

Electric field intensity:

If a charge q will experience a force F in an electric field E then electric field intensity is defined as

$$\vec{E} = \frac{\vec{F}}{q_0}$$

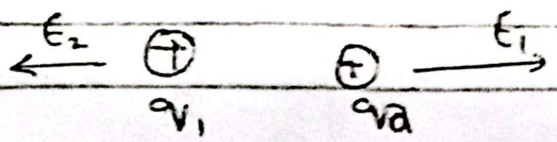
• Vector Quantity • NC^{-1}

• direction is always in direction of force

q_0 must be small charge so that it will not disturb field produced by source charge

$$E = \lim_{q_0 \to 0} \frac{F}{q_0}$$

Charge $\rightleftharpoons$ Field $\rightleftharpoons$ Charge



$$F_{21} = q_2 E_1$$

$$F_{12} = q_1 E_2$$

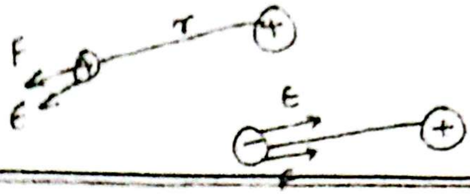
Electric field due to point Charge:

Consider a positive test charge q_0 at a distance "r" from a point charge "q" magnitude. The coulomb force is given

$$F = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_0|q|}{r^2}$$

Magnitude of electric field at site of charge q_0 will be

$$E = \frac{F}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$



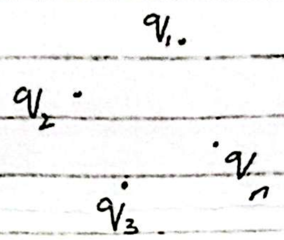
* F and E will point outward if q is +ve.
* F and E will point Inward if q is -ve.

Vector form

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \times \frac{q}{r^2} \times \hat{r}$$

Electric Field due to many Charges (N-Charges)

Let us consider $q_1, q_2 \ldots q_n$ no of charges and their associated electric field $E_1, E_2 \ldots E_N$ respectively.



We want to calculate electric field due to N-charges. So, we will take following procedure.

(i) Calculate E_n due to each "n" Charge = Assuming absence of ~~other~~ charge

(ii) Calculate total "E" by taking vector sum of intensities of individual charge.

$$E = E_1 + E_2 + \cdots E_N$$

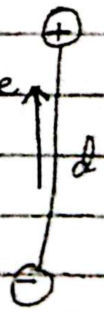
$$E = \Sigma E_N$$

This application of super position.

Electric field due to a Dipole:

* Dipole:

A positive and negative charge having same magnitude and separated by a constant distance d is called dipole.



$\overset{\delta+}{H} - \overset{\delta-}{Cl}$

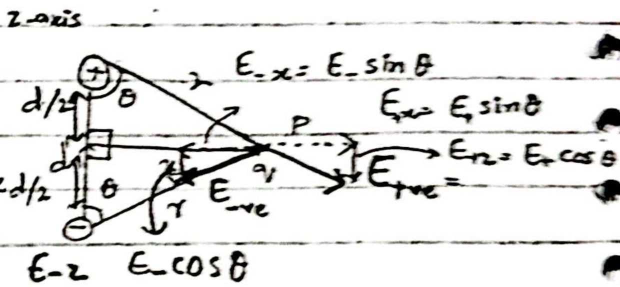
Dipole Moment :

Product of magnitude of charge and separation b/w them is called dipole moment

$$p = qd$$

* It is a vector quantity. * Its direction is always from negative to positive charge.

Consider a positive and a negative charge having same magnitude and constant distance d b/w them, constitute a Dipole.



We want to calculate electric field at point "P" due to dipole.

Draw a perpendicular bisector from line joining the charges to point "P" at distance "x".

Since point "P" is at equal distance from positive and negative charge. So, the magnitude of electric field at point "P" will be same for both charges

$$E_{+ve} = E_{-ve} = \frac{1}{4\pi\varepsilon_0} \times \frac{q}{r^2}$$

Total E :

As electric field follows the principle of super position so total electric field will be

$$E = E_{+ve} + E_{-ve}$$

x- Component of E :

$E_{+}\sin\theta$ and $E_{-}\sin\theta$ are equal in magnitude

But opposite in direction. So, they will cancel each other.

$$E = E_+ \sin\theta + (-E_- \sin\theta)$$

$$E_+ = E_-$$

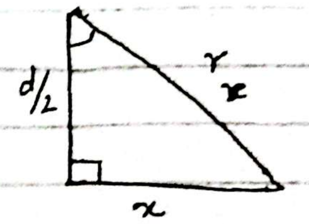
$$= E_+ \cancel{\sin\theta} - E_+ \cancel{\sin\theta}$$

$$E = 0$$

z- Component of E:

$$E = E_+ \cos\theta + E_- \cos\theta$$

$$E = 2E_+ \cos\theta$$



$$E = \cancel{2}\left(\frac{1}{4\pi\varepsilon_0} \times \frac{q}{r^2}\right)\left(\frac{d/\cancel{2}}{r}\right)$$

$$r = \sqrt{x^2 + \left(\frac{d}{2}\right)^2}$$

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{qd}{r^3}$$

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{qd}{\left(x^2 + \left(\frac{d}{2}\right)^2\right)^{3/2}}$$

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{P}{\left(x^2 + \left(\frac{d}{2}\right)^2\right)^{3/2}} \qquad \therefore P = qd$$

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{P}{x^3\left(1 + \left(\frac{d}{2x}\right)^2\right)^{3/2}}$$

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{P}{x^3} \times \left(1 + \left(\frac{d}{2x}\right)^2\right)^{-3/2}$$

Apply Binomial expansion-

$$(1+x)^n = 1 + nx + \cdots\cdots$$

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{P}{x^3} \left[1 + \left(\frac{-3}{2}\right)\left(\frac{d}{2n}\right)^2 + \cdots \right]$$

Case 1 :

$[x >>> d]$

$d \simeq 0$

$$E \neq \frac{1}{4\pi\varepsilon_0} \cdot \frac{P}{x^3}(1+0)$$

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{P}{x^3}$$